

Import Data using Transform Maps (Spreadsheet)

PROJECT DESIGN PHASE

. High-Level Architecture & Workflow Design

The project follows a structured, multi-tier data pipeline model that moves unstructured spreadsheet inputs into a structured enterprise database, followed by business intelligence visualization:

- **Ingestion Layer:** Raw external data files (.xlsx) are uploaded via the Load Data module into an intermediate staging area known as an Import Set table (u_employee_import).
- **Transformation Layer:** A Transform Map (Sample Spreadsheet Import) acts as the bridge, applying field mapping rules and de-duplication logic (Coalesce) to transition data securely into the production target table (u_employee_test).
- **Presentation Layer:** Data is extracted via ServiceNow Reporting tools and organized into widget-based Dashboards (Employee Analytics Dashboard) for administrative oversight.

2. Database Schema & Data Model Design

- **Staging Table (u_employee_import):**
 - *Purpose:* Temporarily stores raw rows and columns from imported spreadsheets before processing.
 - *Fields:* Automatically populated source fields mapping directly to file headers.
- **Target Table (u_employee_test):**
 - *Purpose:* The production custom table holding clean, verified employee master data.
 - *Schema Specifications:*
 - Employee ID (Type: String, Display: false) — Design Note: Designated as the primary unique key for Coalesce mapping.
 - Employee Name (Type: String)
 - Email (Type: String)
 - Department (Type: String) — Used for category grouping in analytics.

- Location (Type: String) — Used for geographic grouping in reporting.

3. Data Flow & Mapping Logic Design

- **Field Mapping Strategy:**
 - Source fields (u_name, u_employee_id, u_department, u_location, u_email) are mapped directly to their corresponding target attributes using the Mapping Assist utility.
- **De-duplication Engine (Coalesce Design):**
 - Coalesce is toggled to true specifically on the u_employee_id field mapping.
 - *Execution Logic:* When an import runs, the system scans the target table for a matching Employee ID. If found, it updates existing records; if absent, it inserts a new entry, preventing redundancy.

4. Reporting & Dashboard Design Structure

- **Report Components:**
 - *Report 1:* Employees by Department (Pie Chart visualization grouped by the Department field).
 - *Report 2:* Employees by Location (Bar Chart visualization grouped by the Location field).
 - *Report 3:* Employee List Report (Tabular list view displaying ID, Name, Email, Department, and Location).
- **Dashboard Design & Access Control:**
 - *Container:* Centralized under Employee Analytics Dashboard.
 - *Security Design:* Requires Access Control List (ACL) modification (pa_dashboards) with Admin Overrides enabled to allow structured layout configuration, alongside role-based sharing parameters for targeted user access.